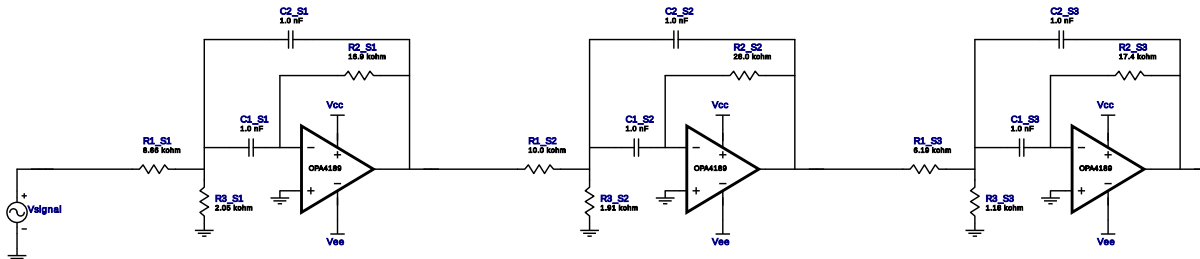


Filter Design Report

Design : Bandpass Filter - 6th order Bessel
Design ID: 21



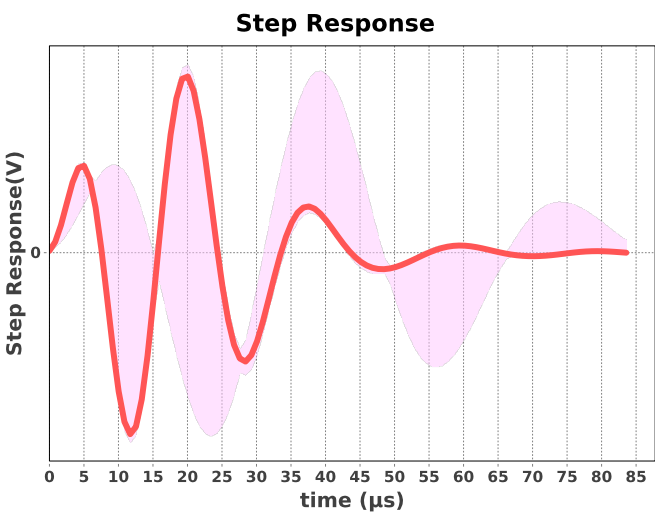
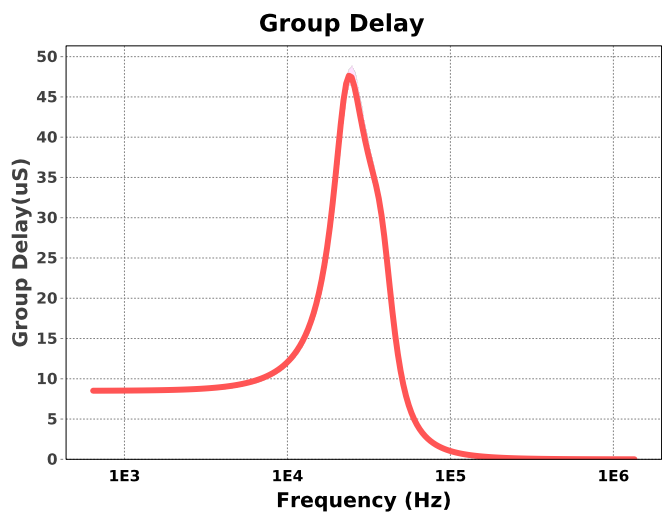
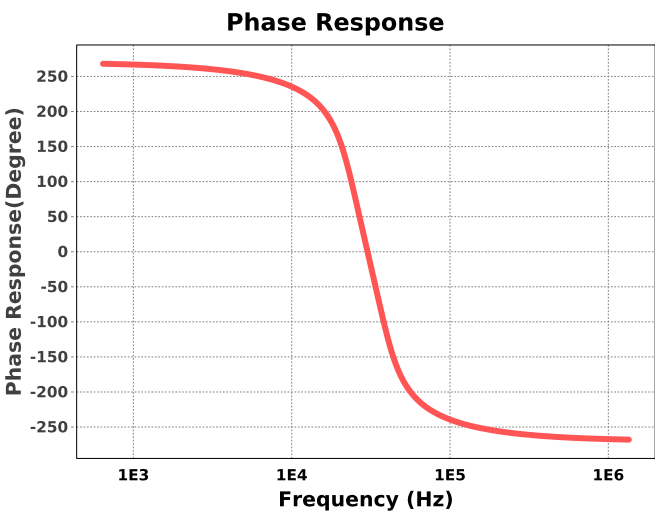
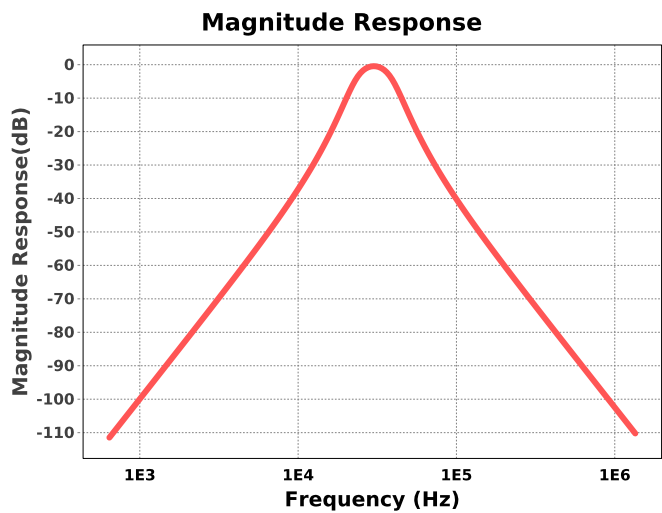
Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	OPA4189	GbwTyp= 14MHz VccMax= 36V VccMin= 4.5V	1
2.	A1_S2	Texas Instruments Inc.	OPA4189	GbwTyp= 14MHz VccMax= 36V VccMin= 4.5V	1
3.	A1_S3	Texas Instruments Inc.	OPA4189	GbwTyp= 14MHz VccMax= 36V VccMin= 4.5V	1
4.	C1_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
5.	C1_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
6.	C1_S3	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
7.	C2_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
8.	C2_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
9.	C2_S3	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
10.	R1_S1	Generic	Ideal	Res= 8660.0ohm Tolerance= 1%	1
11.	R1_S2	Generic	Ideal	Res= 10000.0ohm Tolerance= 1%	1
12.	R1_S3	Generic	Ideal	Res= 6190.0ohm Tolerance= 1%	1
13.	R2_S1	Generic	Ideal	Res= 16900.0ohm Tolerance= 1%	1
14.	R2_S2	Generic	Ideal	Res= 28000.0ohm Tolerance= 1%	1
15.	R2_S3	Generic	Ideal	Res= 17400.0ohm Tolerance= 1%	1
16.	R3_S1	Generic	Ideal	Res= 2050.0ohm Tolerance= 1%	1

#	Name	Manufacturer	Part Number	Properties	Qty
17.	R3_S2	Generic	Ideal	Res= 1910.0ohm Tolerance= 1%	1
18.	R3_S3	Generic	Ideal	Res= 1180.0ohm Tolerance= 1%	1

Sensitivity Analysis

#	Name	Series	Tolerance
1.	Cap	E48	2%
2.	Res	E96	1%



Design Inputs

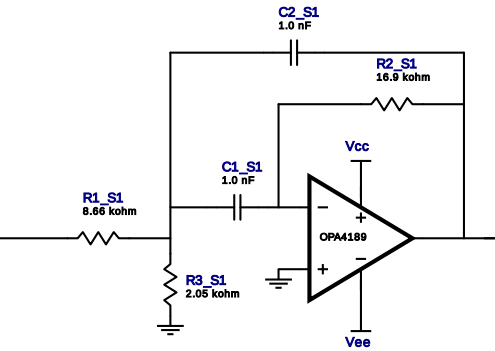
#	Name	Value	Description
1.	FilterType	bandpass	
2.	FilterResponse	Bessel	
3.	FilterOrder	6.0	
4.	FilterTopology	Multiple Feedback	
5.	NumberOfStages	3.0	
6.	CenterFrequency	30.0 k	
7.	StopbandAttenuation	-51.146	
8.	PassbandBandwidth	14.0 k	
9.	StopbandBandwidth	140.0 k	
10.	Gain	1.0	
11.	DualSupply	+/-5.00 V	Power supply(s) to active chips
12.	ResistorTolerance	E96	Resistor series - 1% Passive resistor tolerance
13.	CapacitorTolerance	E48	Capacitor series - 2% Passive capacitor tolerance

Design Assistance

1. **OPA4189** Product Folder : <http://www.ti.com/product/OPA4189> : contains the data sheet and other resources.

Filter Stage :1

Cutoff Frequency 30.07 kHz
Min GBW Req'd 4.844 MHz
Stage Gain 975.751 mV/V
Stage Q 1.597
Stage Topology Multiple Feedback

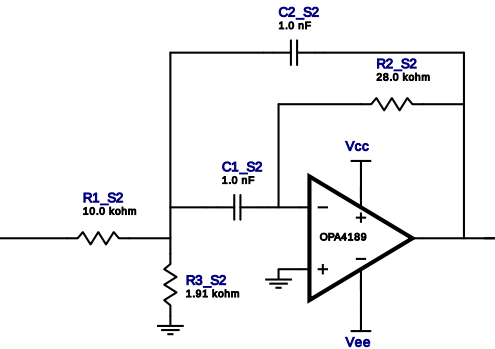


Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	OPA4189	GbwTyp= 14MHz VccMax= 36V VccMin= 4.5V	1
2.	C1_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
3.	C2_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
4.	R1_S1	Generic	Ideal	Res= 8660.0ohm Tolerance= 1%	1
5.	R2_S1	Generic	Ideal	Res= 16900.0ohm Tolerance= 1%	1
6.	R3_S1	Generic	Ideal	Res= 2050.0ohm Tolerance= 1%	1

Filter Stage :2

Cutoff Frequency 23.751 kHz
Min GBW Req'd 7.046 MHz
Stage Gain 1.4 V/V
Stage Q 2.089
Stage Topology Multiple Feedback

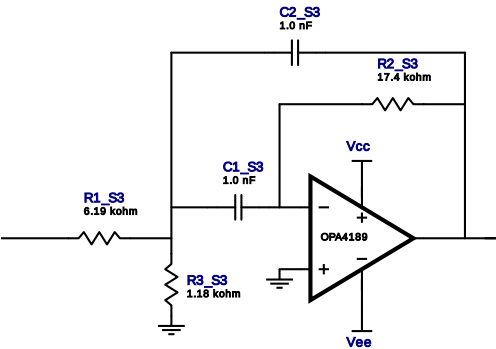


Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S2	Texas Instruments Inc.	OPA4189	GbwTyp= 14MHz VccMax= 36V VccMin= 4.5V	1
2.	C1_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
3.	C2_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
4.	R1_S2	Generic	Ideal	Res= 10000.0ohm Tolerance= 1%	1
5.	R2_S2	Generic	Ideal	Res= 28000.0ohm Tolerance= 1%	1
6.	R3_S2	Generic	Ideal	Res= 1910.0ohm Tolerance= 1%	1

Filter Stage :3

Cutoff Frequency 38.326 kHz
Min GBW Req'd 11.355 MHz
Stage Gain 1.405 V/V
Stage Q 2.095
Stage Topology Multiple Feedback



Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S3	Texas Instruments Inc.	OPA4189	GbwTyp= 14MHz VccMax= 36V VccMin= 4.5V	1
2.	C1_S3	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
3.	C2_S3	Generic	Ideal	Cap= 1.0 nF Tolerance= 2.0 %	1
4.	R1_S3	Generic	Ideal	Res= 6190.0ohm Tolerance= 1%	1
5.	R2_S3	Generic	Ideal	Res= 17400.0ohm Tolerance= 1%	1
6.	R3_S3	Generic	Ideal	Res= 1180.0ohm Tolerance= 1%	1

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